

# Test on Binary File Handling

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## Practice Question: Binary File CRUD System (Python)

### Problem Statement

Design and implement a menu-driven Python program to manage **student records** using a **binary file**.

The program should store data in a binary file using modules like pickle.

### Requirements

#### 1. Create Record (C)

- Input:
  - Roll Number (int)
  - Name (string)
- Marks (float)
- Store the record in a binary file (students.dat)

#### 2. Read / Display Records

- Display all student records stored in the file

#### 3. Update Record (U)

- Search student by roll number
- Modify:
  - Name or Marks

#### 4. Delete Record (D)

- Delete a student record using roll number

### Constraints

- Use **binary file handling** (rb, wb, ab)
- Use **pickle module**
- Use **menu-driven approach**
- Implement **exception handling**
- Perform **data validation**

### Menu

1. Add Student Record
2. Display All Records
3. Search Student
4. Update Record

## 5. Delete Record

## 6. Exit

Solution:

```
import pickle
import os
FILE = "FileHandling/DataFiles/students.dat"
# ----- CREATE -----
def add_record():
    try:
        f = open(FILE, "ab")
        roll = int(input("Enter Roll No: "))
        name = input("Enter Name: ")
        marks = float(input("Enter Marks: "))
        record = [roll, name, marks]
        pickle.dump(record, f)
        print("Record added successfully!")
        f.close()
    except Exception as e:
        print("Error:", e)

# ----- READ -----
def display_records():
    try:
        f = open(FILE, "rb")
        print("\nStudent Records:")
        while True:
            record = pickle.load(f)
            print(record)
        except EOFError:
            f.close()
    except FileNotFoundError:
        print("No records found!")

# ----- SEARCH -----
def search_record():
    try:
        f = open(FILE, "rb")
        roll = int(input("Enter Roll No to
search: "))
        found = False
        while True:
            record = pickle.load(f)
            if record[0] == roll:
```

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        print("Record Found:", record)
        found = True
        break
    except EOFError:
        if not found:
            print("Record not found!")
            f.close()

# ----- UPDATE -----
def update_record():
    try:
        f = open(FILE, "rb")
        temp =
open("FileHandling/DataFiles/temp.dat", "wb")
        roll = int(input("Enter Roll No to
update: "))
        found = False
        while True:
            record = pickle.load(f)
            if record[0] == roll:
                print("Old Record:", record)
                record[1] = input("Enter new
name: ")
                record[2] = float(input("Enter
new marks: "))
                found = True
                pickle.dump(record, temp)
        except EOFError:
            f.close()
            temp.close()
            if found:
                os.remove(FILE)
                os.rename("FileHandling/DataFiles/t
emp.dat", FILE)
                print("Record updated!")
            else:
                os.remove("FileHandling/DataFiles/t
emp.dat")
                print("Record not found!")

# ----- DELETE -----
def delete_record():
    try:
        f = open(FILE, "rb")
        temp =

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open("FileHandling/DataFiles/temp.dat", "wb")
    roll = int(input("Enter Roll No to
delete: "))
    found = False
    while True:
        record = pickle.load(f)
        if record[0] != roll:
            pickle.dump(record, temp)
        else:
            found = True
    except EOFError:
        f.close()
        temp.close()
        if found:
            os.remove(FILE)
            os.rename("FileHandling/DataFiles/t
emp.dat", FILE)
            print("Record deleted!")
        else:
            os.remove("FileHandling/DataFiles/t
emp.dat")
            print("Record not found!")

# ----- MENU -----
def main():
    while True:
        print("\n--- Student Management
System ---")
        print("1. Add Record")
        print("2. Display Records")
        print("3. Search Record")
        print("4. Update Record")
        print("5. Delete Record")
        print("6. Exit")
        choice = int(input("Enter choice: "))
        if choice == 1:
            add_record()
        elif choice == 2:
            display_records()
        elif choice == 3:
            search_record()
        elif choice == 4:
            update_record()
        elif choice == 5:
            delete_record()

```

```
elif choice == 6:  
    break  
else:  
    print("Invalid choice!")  
  
if __name__ == "__main__":  
    main()
```